

# SPIKE GATE PHASE 2 IMPROVEMENT OPTIONS

Prioritised route from conservative diagnostic to validated masking workflow

Prepared: 20 August 2026

## RECOMMENDED DIRECTION

Retain MTOjects Phase 2 as the diagnostic baseline. Improve gate credibility and candidate matching first, then add local recovery only for credible unmatched candidates. Do not broaden the global mask to recover misses.

### 1. PRIORITISED IMPROVEMENTS

1. Gate credibility model: score amplitude, side-drop, width, residual compactness, multi-width persistence and 2-D residual agreement.
2. Native-coordinate target and support: use one traceable coordinate system and conserved denominators.
3. Candidate-specific association: report matched, unmatched, split and merged candidates; optimise candidate-level recall and precision.
4. Morphology protection: penalise annular coherence, azimuthal span, arm-like elongation and low-frequency galaxy overlap.
5. Two-stage local recovery: search a small native window for credible misses under tighter area and morphology caps.
6. Manual labelled validation: label approximately 20-30 galaxies including NGC4020, NGC4532 and NGC5236.
7. Stratified cross-validation: balance image scale, inclination, morphology, gate count and foreground density.
8. Runtime controls: cap raw components, enforce time and memory limits and checkpoint/resume.

### 2. STAGED PROGRAMME

Stage A: build gate-credibility labels and candidate-level matching diagnostics.

Stage B: implement morphology protection and native-coordinate accounting.

Stage C: run stratified four-fold optimisation on the labelled subset.

Stage D: add local recovery for credible unmatched gates and re-audit misses.

Stage E: approve a new 182-galaxy batch only if quantitative and visual criteria pass.

### 3. ACCEPTANCE CRITERIA

Mean credible-candidate recovery at least 0.70.

Candidate detection at least 0.75.

No unexplained zero detections for high-credibility candidates.

Mean protected galaxy loss no more than 0.01.

Mean and maximum native masked fractions no more than 0.02 and 0.05.

No confirmed coherent structure removal in the labelled audit.

Stable behaviour across four stratified folds.

#### 4. DECISION RULE

Advance to production-style use only when candidate recovery improves without exceeding morphology-protection and area limits. If local recovery increases galaxy loss, retain the conservative baseline and route uncertain cases to manual review.